

1. Matched records

Member M
account | Pro | EU
hidden email:
mira@synthx.invalid

Matched nonmember N
account | Pro | EU
hidden email:
niko@synthx.invalid

**Only M is inserted
into training**

2. Same audit protocol

Same high-cue prompt
plan=Pro, region=EU
Complete recovery_email:

**Same cue band
same budget B=5
same match rule**

Budgeted scoring
extracted = any of B completions
matches the hidden target field

3. Exposure-specific evidence

Member completion
mira@synthx.invalid
match = 1

Nonmember completion
plausible email
match = 0

Aggregate over matched pairs

$$\Delta_{\text{mem}} = r_{\text{mem}} - r_{\text{non}}$$

Behavioral readout
C3-fuzzy-5x, high cue, B=5
member 39.5% | nonmember 32.9%
lift +6.5 pp

Provenance diagnostic
rank same-family candidates
by gradient similarity

If extracted content is training-linked,
the inserted record or fuzzy cluster
should rank near the top.

The nonmember path estimates prompt-only reconstruction; lift isolates the extra extraction probability attributable to training exposure.